

# IBM Essential PYQ Practice Pack

Service Based IT / IBM

## Study Hub Premium Placement Pack

This is original study material for preparation. It is not a copied official or proprietary company paper.

## Premium Study System

- Track: IBM.
- Pack type: PYQ / PYQ Practice.
- Target outcome: selection-ready confidence for Graduate Engineer / Analyst.
- Core preparation areas: quant, reasoning, verbal, programming logic, CS fundamentals, communication, HR answers.
- Use this pack like a mini-book: concept, table, example, practice, analysis, revision.
- Every topic should produce proof: solved questions, mock score, mistake note, interview answer, or resume evidence.

## Output Quality Rules

- Read the concept in your own words, then solve without looking at notes.
- Convert every important idea into a comparison table, checklist, or one-line rule.
- Mark important traps as Common Mistake before the mock, not after losing marks.
- Keep a PYQ-style log: question pattern, topic, time taken, mistake reason, retest date.
- Use simple English; add Hindi/Hinglish meaning in brackets when it helps recall.

## Priority Matrix

| Area | Premium Standard | Proof Before Moving On |  
| Aptitude | Formula, shortcut, accuracy, time control | 25 mixed questions with 70%+ accuracy |  
| Programming Logic | Loops, arrays, strings, SQL basics | Trace output and write simple code cleanly |  
| Project | Problem, architecture, your contribution, tradeoff | 2-minute story plus deep-dive answers |  
| Communication | Crisp intro, structured HR stories, calm tone | 5 recorded answers reviewed once |  
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## Exam Tip

- Service assessments reward speed with accuracy. Attempt high-confidence sections first and protect time for easy marks.

## Common Mistake

- Spending too long on one aptitude question reduces total score. Set a strict skip rule and revisit only after easy wins.

## Original PYQ Style Set

- This is original practice inspired by common placement patterns, not a copied company paper.

## Original PYQ Style Practice

- This set is designed for IBM style campus preparation: aptitude speed, programming logic, SQL, CS basics, communication, and HR clarity.

## Section A: Quantitative Aptitude

- Q1. A candidate solves 60% of a set in 36 minutes. Estimate the total time for the full set and explain the shortcut.
- Q2. A, B, and C invest in ratio 3:4:5. After 8 months A doubles investment. Find final profit

ratio after 12 months.

- Q3. A pipe fills a tank in 12 hours and another empties it in 18 hours. If both are open, find net time and the faster shortcut.
- Q4. In a class, average marks rise from 64 to 67 when one student scoring 28 is replaced. Find the new student score for class strength 40.

## Section B: Reasoning and Verbal

- Q5. Five candidates sit in a row. A is left of B, C is not at an end, and D is right of E. Build two possible arrangements.
- Q6. Complete the series: 3, 8, 18, 38, 78, ?. Write the pattern in one sentence.
- Q7. Pick the sentence with the clearest business communication and rewrite the weakest option.
- Q8. Write a 90-word email asking HR to reschedule an interview politely and professionally.

## Section C: Programming Logic and Coding

- Q9. Trace a loop that updates sum only when i is divisible by 3 or 5. State final output for n = 20.
- Q10. Given an array, return the second highest distinct value. Include edge cases for duplicates and negative values.
- Q11. Count frequency of each character in a string and print the first non-repeating character.
- Q12. Write pseudocode to reverse words in a sentence without reversing the characters inside each word.

## Section D: SQL and CS Fundamentals

- Q13. Write a query to find departments with average salary above the company average.
- Q14. Explain primary key vs unique key vs foreign key with one example table.
- Q15. Explain process vs thread, normalization, indexes, HTTP status codes, and OOP inheritance with examples.
- Q16. What happens when you type a URL in browser? Explain DNS, TCP, HTTP, server, and response in simple words.

## Section E: HR and Communication

- Q17. Tell me about yourself for a fresher role at IBM. Keep the answer under 60 seconds.
- Q18. Why IBM? Connect learning, role fit, adaptability, and your preparation evidence.
- Q19. Tell me about a failure. Use STAR format and end with what changed in your behavior.
- Q20. Explain your best project to a non-technical interviewer in 90 seconds.

## Model Hints

- Q1 hint: if 60% takes 36 minutes, 100% takes  $36 \times 100 / 60 = 60$  minutes.
- Q3 hint: net work per hour =  $1/12 - 1/18 = 1/36$ , so tank fills in 36 hours.
- Q10 hint: track largest and second largest distinct values; handle fewer than two distinct values.
- Q13 hint: use GROUP BY department and HAVING AVG(salary) > (SELECT AVG(salary) FROM employees).

## Solution Checklist

- Write assumptions before solving aptitude.
- Show dry run for every coding answer.
- Use STAR format for HR answers.
- Keep final answer short, structured, and interview-friendly.
- After the set, classify mistakes as concept gap, formula gap, reading error, time trap, or panic.

## How To Review

- Mark every question as solved, guessed, skipped, or wrong.
- Rewrite the idea behind every wrong question in one sentence.

- Re-attempt wrong questions after 24 hours and again after 7 days.

## **PYQ-Style Attempt Protocol**

- Step 1: identify topic and difficulty before solving.
- Step 2: write the first approach, even if it is brute force or long method.
- Step 3: optimize only after the basic idea is correct.
- Step 4: dry run one normal case and one edge case.
- Step 5: add answer, time, mistake type, and retest date to the log.

## **Pattern References**

- Common placement patterns: percentage change, ratio split, time-work efficiency, seating arrangement, sentence correction, loop tracing, SQL grouping.

## **Answer Review Table**

| Mark | Meaning | Next Action |  
 | Solved | correct under timer | revise after 7 days |  
 | Guessed | answer came without method | solve two similar questions |  
 | Wrong | concept or execution gap | write mistake rule and reattempt tomorrow |  
 | Skipped | time or confidence issue | learn shortcut or move to lower difficulty |

## **Execution Dashboard**

| Metric | Target | Review Frequency |  
 | Aptitude Sets | 5 timed sets per week | every Sunday |  
 | CS Core Notes | 4 topics per week | twice a week |  
 | Mock Tests | 1-2 per week | same day analysis |  
 | Interview Practice | 5 answers per week | record and review |

## **Weekly Review Ritual**

- Pick the top 5 mistakes, rewrite the correct rule, and solve one similar question.
- Remove one weak resume line or prepare one stronger project answer.
- Decide the next week from evidence, not mood.

## **Final Self Audit**

| Score | Meaning | Action |  
 | 0-40 | reading stage | finish basics and easy questions |  
 | 41-70 | practice stage | increase timed mocks and mistake review |  
 | 71-85 | interview-ready | polish project, HR, and weak topics |  
 | 86-100 | selection-ready | maintain revision and avoid overloading new material |

## **Source Note**

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